

Keshav Ganapathy

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EDUCATION

Georgia Institute of Technology

M.S. in Machine Learning

Aug. 2025 - Present

GPA: 4.00

University of Maryland – College Park

Aug. 2021 – Dec 2024

B.S. in Computer Science (Machine Learning, with Honors) & Mathematics, Minor: Computational Finance

- **Relevant Coursework:** Advanced Data Structures, Advanced Algorithms, Machine Learning, Real Analysis, Data Science
- **Activities:** Teaching Assistant (CMSC351: Algorithms), BigThinkAI (500+ Members, President), Club Pickleball

WORK EXPERIENCE

Capital One

McLean, VA

Software Engineer, Generative AI Platform

June 2024 – Present

- Engineer on GenAI platform team, responsible for developing and maintaining a one-stop self-serve service to deploy production-bound GenAI applications, ranging from basic inference to multi-turn multi-agent use cases.
- Led a platform-wide migration to **GitHub Enterprise Cloud (GHEC)** across **200+ repositories** that were used by **30+ production-bound use cases**, achieving **zero downtime** and delivering platform-wide enterprise compliance.
- Onboarded **5+** use cases to the GenAI platform, deploying LLM inference and RAG-based solutions to production.
- Managed three interns to design and build a Slack Chatbot for **550+** GenAI customers, using a four-tier system (Ingestion, RAG, Lambda, Slack API), providing instant answers to errors and significantly reducing on-call engineering load.
- Re-engineered the GenAI template test suite with **Jinja**, **JMeter**, and **pytest**, improving runtime by **2x+**, boosting code coverage, saving **15+ developer hours per use case**, while authoring documentation to streamline framework adoption.
- Engineered a user information timeline, bettering customer experience, using **PySpark** on customer data in **AWS S3**, processing over **330M+ daily entries** and improving query speed by **300%** through serialization, caching, and partitioning.
- Machine Learning Intern (2024), Software Engineer (Feb 2025); Certified **AWS Solutions Architect – Associate**

Raytheon

Arlington, VA

Software Engineering Intern, BBN Technologies

May 2023 - August 2023

- Refactored and enhanced an internal tool by leveraging **CesiumJS** and **Python** to display **100+** satellites and their maneuvers from an **SQLite** database, resulting in novel 3D globe, map, and satellite visualizations.
- Implemented and optimized **Orekit** orbital propagation algorithms in **Python** to generate training data for space-object-collision detection machine learning models to accept covariance matrices for more accurate predictions

Cigna

Bloomfield, CT

Software Engineering Intern

May 2022 - March 2023

- Completed full-stack tickets for myPassport, Cigna's main pharmacy benefit management tool for worker's compensation.
- Led **10+ interns** in the development of 4Sight, an internal web app that provides Cigna employees information on in-person attendance. 4Sight **increases employee retention** rate and promotes community amongst employees.

University of Maryland

College Park, MD

Researcher

June 2020 - Aug 2020, June 2021 - Nov 2021

- Published research on author demographics at ICLR and the ML conference review process (NeurIPS Workshop).
- Co-first-authored a highly impactful paper that was presented at the **ICLR 2021 townhall** and selected as a **lightning talk** at the **NeurIPS 2020** Workshop on Navigating the Broader Impacts of AI Research. Papers and talks analyzed and quantified disparities in the review process based on factors such as gender, institutional affiliation, etc.

PROJECTS

GenAI (Element Solutions, Co-Op Fall/Spring 2024) | AWS, FastAPI, React, Ollama, RAG, and ChromaDB

- Built, dockerized, and deployed an ingest and **RAG-based Q&A** service on **AWS EC2** that pulls files from **S3** and fine-tunes **HuggingFace embeddings**, allowing for fast and accurate customer Q&A about Element Solutions' LOBs and past work.

CityWorks (Element Solutions, Co-Op Spring 2022) | AWS, Spring Boot, React, and MariaDB

- Successfully worked in a team of **6+ engineers** using **agile scrum** to create a full stack application using **React**, **AWS**, and **MySQL** to allow **hundreds of citizens** to request various social services (e.g trash pick up)..
- Designed the database schema and developed **RESTful API** endpoints using **Spring Boot** in **Java** for MySQL.

WebPoint Portfolio | HTML, CSS, Pug, SCSS, JavaScript

- Founded WebPoint, a volunteer organization that developed **5** websites for local enterprises, receiving **100+** views per week.

SKILLS

- **Languages:** Python, Java, C/C++/C#, JavaScript, SQL, MATLAB, SASS, OCaml, Ruby, Rust, Assembly (MIPS), $\LaTeX$
- **Frameworks/Libraries:** Langchain, PySpark, Pandas, TensorFlow, Sklearn, Pytorch, Flask, NodeJS, React, Springboot
- **Tools & Technologies:** AWS, Git, Postman, Docker, Jira, Confluence, Maven, MS Office/GSuite